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# ProfessionalBench: Benchmarking AI Agents on Real-World Professional Workflows

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## Abstract

As AI agents handle files, use tools, and produce complex deliverables, evaluating their ability to perform professional work becomes increasingly important. Recent benchmarks adopt workflow-based tasks to improve realism, yet their construction can underrepresent professional work in both breadth and depth. Ad hoc task sampling leaves professional units and workflows uncovered, while *Workflow Compression* omits or pre-resolves responsibilities within the designated role, preventing direct assessment of whether agents can fulfill them independently. To capture real professional workflows, we introduce **ProfessionalBench**, spanning 30 subdomains across finance, law, and healthcare. Experts compare tasks against source workflows and use a *Know–Act–Close (KAC)* formulation to preserve interdependent responsibilities for professional judgment, execution, and closure. Expert-reviewed *Dense Rubrics*, aligned with KAC responsibilities, support hybrid verification of requirement completion and deliverability, while trajectory analysis identifies failure mechanisms behind unmet requirements. We evaluate seven frontier agents on ProfessionalBench, all achieving Professional Pass rates below 40% in the main evaluation. Workflow Compression interventions show that pre-resolving responsibilities increases mean scores across three models, with larger average gains from compressing Close (+7.8 points) and Act (+6.0) than Know (+3.7) and 12.2–20.0% of matched pairs transition from failure to Professional Pass under fixed criteria. Trajectory-based failure diagnosis attributes the largest failure shares to Act in finance and law, and Close in healthcare. Together, these findings show that preserving role-specific responsibilities exposes substantial professional headroom and identifies where agents need to improve.

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**Project Page:** https://ProfessionalBench.github.io/
**Date:** October 14, 2026

## 1 Introduction

Large language model (LLM) agents have made substantial progress in long-horizon reasoning, information retrieval, code execution, and tool use, extending their capabilities to increasingly complex real-world work. Evaluation has accordingly shifted toward a more consequential question: *can current agents reliably perform professional work in realistic occupational settings?* Working agent benchmarks increasingly incorporate authentic workspace materials, open-ended deliverables, enterprise software, and domain-specific tools, often involving domain experts in task construction and evaluation design. This shift emphasizes end-to-end workflows with explicit business objectives, professional constraints, and delivery requirements.

Despite this progress, real-world provenance alone does not guarantee a faithful and comprehensive represen-

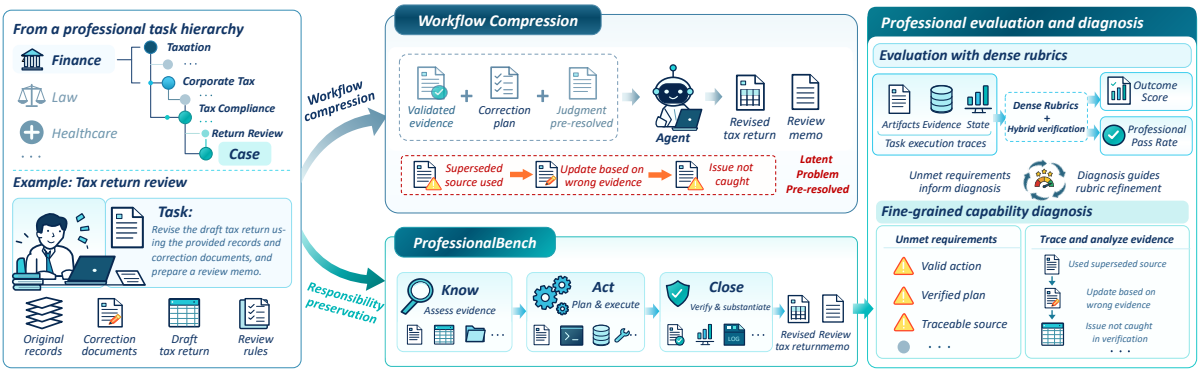



**Figure 1 Position of ProfessionalBench.** We organize tasks hierarchically, preserve Know–Act–Close responsibilities to address Workflow Compression, and enable fine-grained evaluation.

tation of professional work. Simply expanding domain coverage and increasing task counts does not ensure systematic coverage of the professional units and canonical workflows within each domain. More importantly, even when a workflow is included, its instantiated task may fail to preserve the critical responsibilities that a professional would normally undertake independently. During task construction, responsibilities such as evidence assessment, execution planning, and outcome verification may be omitted or pre-resolved through supplied conclusions or verification results. Agents are then required to complete only the remaining work, without independently addressing the corresponding professional problems and their dependencies.

We refer to this omission or pre-resolution of critical professional responsibilities during task instantiation as *Workflow Compression*. Through agent assessment and expert verification, we audit 837 tasks from four publicly available benchmarks. Across these audited collections, the confirmed rate of joint preservation of professional judgment, execution, and closure ranges from only 6.7% to 45.0% (Table 12). Workflow Compression can be appropriate when the goal is to isolate specific capabilities. However, when changes to a task's responsibility scope are not made explicit, performance on compressed tasks may be mistaken for evidence of end-to-end professional competence. Once a responsibility is omitted or pre-resolved, the benchmark can no longer directly assess whether agents can fulfill it independently, a gap that finer-grained evaluation of the remaining work cannot remedy. These limitations motivate the following research question:

**Research Question**

*How can we construct a benchmark that faithfully represents professional work in both breadth, covering the workflows of each domain, and depth, preserving the responsibilities within each workflow?*

To address this question, we introduce **ProfessionalBench**, a benchmark of real-world professional workflows spanning finance, law, and healthcare. Drawing on occupational resources, industry standards, and expert knowledge, we organize task sourcing and construction through a *Domain–Subdomain–Professional Unit–Anchor Workflow* hierarchy, using identified coverage gaps to guide further expansion. At the task level, we characterize professional depth by the extent to which critical responsibilities and their dependencies are preserved. We operationalize this principle through *Know–Act–Close (KAC)*: agents must independently gather evidence and form professional judgments; plan and execute actions that advance the relevant business state; and verify outcomes, address in-scope exceptions, and determine whether the work is complete. Experts compare each task against its source workflow to identify and restore responsibilities that would be omitted or pre-resolved, while preserving its business objective, role boundary, and professional completion criteria.

To assess whether these responsibilities are fulfilled, we construct expert-reviewed *Dense Rubrics* that link KAC duties and functional capabilities to outcome conditions and traceable evidence of professional judgment, execution, and verification. Programmatic and agent-based verifiers inspect deliverables, evidence records, and final business states. *Outcome Score* summarizes satisfaction of responsibility-aligned criteria; *Professional Pass* requires meeting a specific threshold and critical criteria. We evaluate seven frontier model–harness configurations and conduct paired Workflow Compression interventions. Close compression yields the largest average score gain, while primary failures in original runs most often involve Act and Close, suggesting that agents can complete parts of a task yet fail to fulfill execution and closure responsibilities.

Our contributions are threefold:

- **Responsibility-Preserving Benchmark.** We identify *Workflow Compression* in existing benchmarks and introduce **ProfessionalBench**, which preserves professional work in breadth through a domain–unit–workflow hierarchy and in depth through expert-guided Know–Act–Close construction.
- **Responsibility-Aligned Evaluation.** We combine expert-reviewed Dense Rubrics with hybrid verification to assess requirement completion and professional deliverability. Rubric-level localization and trajectory analysis further characterize unmet requirements and failure mechanisms.
- **Professional Capability Analysis.** We systematically evaluate frontier agent frameworks and use *Workflow Compression* interventions to reveal that responsibility preservation affects measured performance, capability profiles, and the observability of professional bottlenecks.

## 2 Related Work

**Professional Task Evaluation.** Recent benchmarks have shifted toward evaluating AI systems on occupationally relevant and open-ended tasks. GDPval [9] studies economically valuable work from industry professionals, while APEX [14], PRBench [1], and XpertBench [7] evaluate professional tasks with expert-defined criteria. ALE [11] further introduces occupational taxonomy and workflow-family organization for systematic task construction. Other efforts explore specific settings, including workplace delegation [6], financial agents [18], and coding-oriented professional workflows [12]. However, existing benchmarks largely rely on task sampling and lack systematic coverage of professional units and workflow responsibilities. ProfessionalBench addresses this gap by constructing an expert-calibrated hierarchy across three domains to guide workflow selection and coverage expansion.

**Long-Horizon Workflow Evaluation.** Interactive benchmarks evaluate whether agents can execute multi-step workflows under changing environments. WebArena [20] and OSWorld [16] introduce realistic browser and desktop environments, while WorkArena [4], AppWorld [13], and TheAgentCompany [17] extend evaluation to enterprise applications and workplace interactions. Recent benchmarks further study complex tool use, such as Toolathlon [5], SaaS-Bench [10], and Workflow-GYM [21]. Nevertheless, longer trajectories do not guarantee faithful professional workflows. ProfessionalBench introduces Workflow Compression analysis and expert-guided Know–Act–Close construction to identify missing responsibilities and dependencies.

**Rubric-Based Evaluation and Diagnosis.** LLM-based evaluation methods, including LLM-as-a-judge [19] and G-Eval [8], enable scalable assessment of open-ended outputs. Professional benchmarks further incorporate expert-designed criteria, such as PRBench [1], XpertBench [7], HealthBench [2], and LexRubric [3]. Beyond final-answer scoring, recent work analyzes intermediate traces and criterion-level failures, including JobBench [6], BigFinanceBench [15], and ALE [11]. However, existing frameworks provide limited diagnosis of unmet requirements and failure causes. ProfessionalBench combines responsibility-aligned dense rubrics with hybrid verification for deliverable assessment and fine-grained failure analysis.

## 3 Overview of ProfessionalBench

We begin by outlining our design principles and choices. We then describe ProfessionalBench's construction, quality-control measures, and some descriptive statistics.

### 3.1 Design Principles

ProfessionalBench is designed to preserve what agents must independently accomplish in professional work and support fine-grained capability evaluation. Its design follows three principles:

- **Professional Responsibility Fidelity.** Tasks must preserve source workflows' critical responsibilities and the dependencies among professional judgment, execution, and closure, rather than omitting or pre-resolving responsibilities that agents should fulfill autonomously.
- **High-Resolution Verifiability.** Professional responsibilities must be evaluated against explicit acceptance criteria using observable evidence from artifacts and business states. Evaluation should localize unmet

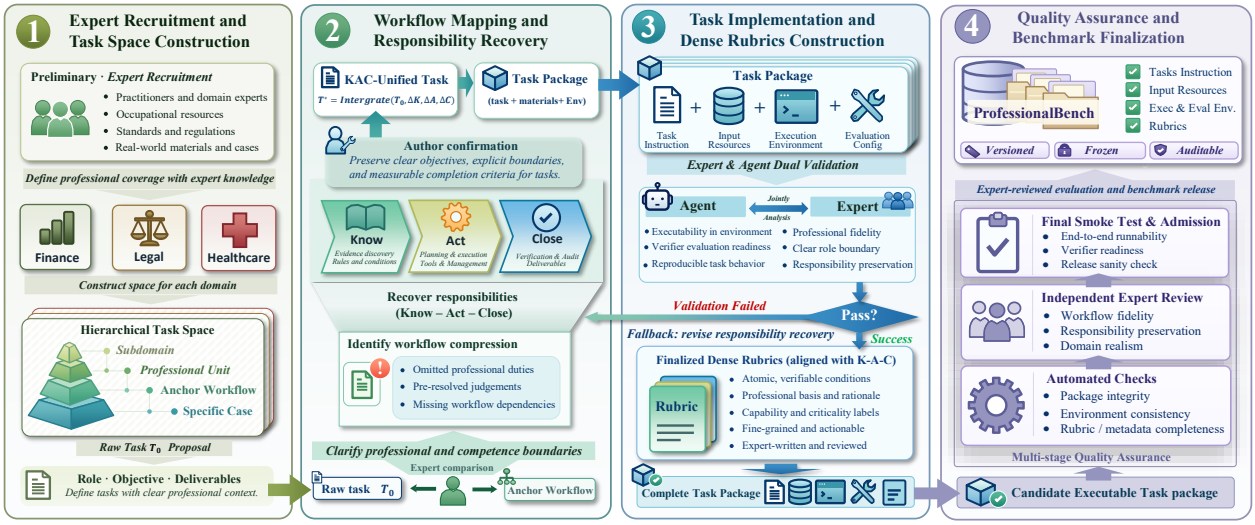



**Figure 2 Construction pipeline of ProfessionalBench.** The pipeline transforms authentic professional workflows into responsibility-preserving, executable, and verifiable KAC-Unified Tasks through expert-guided construction and quality assurance with workflow fidelity preservation.

requirements across responsibilities and capabilities while distinguishing completion from professional deliverability and actionable capability gaps.

- **Structured Professional Coverage.** Professional work is organized around recurring responsibilities and workflows rather than isolated prompts. ProfessionalBench constructs an expert-calibrated task space and selects representative Workflows to achieve systematic coverage across domains and Professional Units.

Guided by these principles, we formalize responsibility-preserving tasks (Sec. 3.2), describe their construction and evaluation (Secs. 3.3-3.5), and summarize the resulting benchmark (Sec. 3.4).

## 3.2 Responsibility-Preserving Workflow Formulation

To operationalize Professional Responsibility Fidelity, ProfessionalBench adopts the *Anchor Workflow* as the basic unit of task formulation: a self-contained unit of professional work with a designated role, business objective, and completion criteria. Relative to the designated role and task boundary, *Workflow Compression* occurs when task construction omits or pre-resolves critical responsibilities that agents should otherwise fulfill.

**Know--Act--Close Responsibility Criteria.** To translate responsibility preservation into task-design criteria, domain experts identify the responsibilities of each Anchor Workflow across three coupled stages:

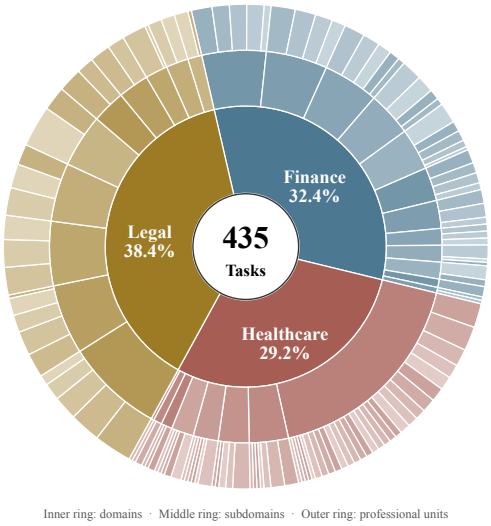
- *Know: Form reliable judgments from evidence.* The agent should independently discover and assess evidence, forming evidence-grounded judgments rather than relying on pre-resolved hints.
- *Act: Advance professional execution.* The agent should apply its judgments through execution, adapting to intermediate results and exceptions rather than following pre-specified procedures.
- *Close: Verify and substantiate completion.* The agent should verify outcomes against completion criteria, addressing in-scope discrepancies and substantiating completion with traceable evidence.

Know, Act, and Close are not independent subtasks or a prescribed one-pass sequence. Judgments formed during Know must guide Act, while Close verifies the resulting outcomes and may trigger renewed evidence assessment or corrective action. A responsibility-preserving task must retain these dependencies around a shared professional objective.

## 3.3 Expert Construction and Quality Assurance

As shown in Figure 2, ProfessionalBench employs a four-stage, expert-led, model-assisted pipeline to transform authentic professional tasks into responsibility-preserving, executable evaluation tasks.

**Table 1** Domain-wise composition and coverage density of ProfessionalBench. The sunburst (left) shows the domain-specific taxonomy; all taxonomy counts are computed within domains, and similarly named nodes in different domains are treated as distinct categories.




| Statistic | Healthcare | Finance | Legal | Total |
|---|---|---|---|---|
| *Benchmark scale* | | | | |
| Subdomains | 8 | 12 | 10 | 30 |
| Professional Units | 58 | 41 | 28 | 127 |
| Anchor Workflows | 113 | 108 | 134 | 355 |
| Selected Tasks | 127 | 141 | 167 | 435 |
| Task Share | 29.2% | 32.4% | 38.4% | 100.0% |
| *Coverage density* | | | | |
| Tasks per Subdomain | 15.9 | 11.8 | 16.7 | 14.5 |
| Tasks per Professional Unit | 2.2 | 3.4 | 6.0 | 3.4 |
| Tasks per Anchor Workflow | 1.12 | 1.31 | 1.25 | 1.23 |
| Professional Units per Subdomain | 7.3 | 3.4 | 2.8 | 4.2 |
| Anchor Workflows per Professional Unit | 1.9 | 2.6 | 4.8 | 2.8 |

**Stage 1: Task-Space Planning and Expert Sourcing.** ProfessionalBench first constructs a hierarchical professional task space spanning finance, law, and healthcare. Drawing on occupational resources (e.g., O*NET activities), industry standards, and expert input, we organize professional work into *30 subdomains* and *127 Professional Units*. Domain experts define representative *Anchor Workflows* for each unit, specifying responsible roles, business objectives, and independently verifiable completion criteria. Based on this task space, practitioners contribute raw tasks $T_0$ together with supporting professional materials, including relevant records, standards, regulations, and practice guidelines, which provide the basis for subsequent responsibility analysis.

**Stage 2: Responsibility Recovery and Task Instantiation.** Each raw task $T_0$ is reviewed against its source Anchor Workflow $W$ by domain experts who have not participated in task creation. Experts identify omitted or pre-resolved responsibilities as KAC gaps $(\Delta_K, \Delta_A, \Delta_C)$, and propose recovery operations with their professional basis and dependency relationships. After review and confirmation, these modifications are integrated into the final task $T^* = \text{Integrate}(T_0, \Delta_K, \Delta_A, \Delta_C)$. The resulting $T^*$ is defined as a *KAC-Unified Task*, preserving the original workflow objective, responsibility boundary, and completion criteria while restoring critical professional responsibilities. Each task is then packaged with standardized metadata and execution materials for subsequent environment construction and evaluation.

**Stage 3: Task Implementation and Dense Rubrics Construction.** KAC-Unified Tasks are instantiated as candidate task packages containing task instructions, input resources, a Docker execution environment, and evaluation configurations. Domain experts translate retained responsibilities into Dense Rubrics with atomic acceptance conditions, expert-assigned weights, and observable verification targets, including required work records and final deliverables. Agent-assisted validation checks executability and evaluation readiness, while experts assess professional fidelity, role boundaries, and responsibility preservation. Task-design failures are returned to Stage 2 for revision.

**Stage 4: Quality Assurance and Benchmark Finalization.** Candidate packages undergo automated integrity and consistency checks, independent expert review, and final end-to-end smoke tests. These checks assess package completeness, workflow fidelity, rubric alignment, and verifier readiness. Packages failing a check are returned for revision and rechecked before admission. Accepted tasks, environments, and evaluation configurations are versioned and frozen for benchmark release.

**Table 2 Main results on ProfessionalBench.** We evaluate seven frontier agents on 90 tasks, with 30 tasks per domain. Score, Pass, Delay, and Cost denote Outcome Score, Professional Pass Rate, average execution time (min), and cost (USD) per task, respectively. Overall results aggregate all 90 tasks. Best and second-best Score and Pass results are bolded and underlined, respectively.

| Model | Finance | | | | Law | | | | Healthcare | | | | Overall | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | Score (0–100) | Pass (%) | Delay (min) | Cost ($) | Score (0–100) | Pass (%) | Delay (min) | Cost ($) | Score (0–100) | Pass (%) | Delay (min) | Cost ($) | Score (0–100) | Pass (%) | Delay (min) | Cost ($) |
| GPT-5.6 Sol | <u>60.83</u> | **36.7** | 5.38 | 1.78 | **63.11** | **30.0** | 13.32 | 4.46 | **66.98** | <u>33.3</u> | 6.58 | 1.76 | **63.64** | **33.3** | 8.43 | 2.67 |
| Claude Opus 5 | **63.57** | <u>33.3</u> | 14.78 | 2.68 | <u>60.74</u> | <u>26.7</u> | 31.35 | 4.49 | <u>64.37</u> | **36.7** | 15.95 | 2.86 | <u>62.89</u> | <u>32.2</u> | 20.69 | 3.34 |
| Gemini 3.8 Flash | 42.10 | 13.3 | 9.78 | 0.81 | 46.53 | 10.0 | 12.23 | 0.84 | 51.14 | 16.7 | 13.80 | 0.71 | 46.59 | 13.3 | 11.94 | 0.79 |
| DeepSeek-V4.1-Flash | 52.04 | 6.7 | 15.22 | 0.25 | 52.04 | 13.3 | 30.93 | 0.79 | 55.05 | 23.3 | 11.10 | 0.26 | 53.04 | 14.4 | 19.08 | 0.43 |
| GLM-5.2 | 50.16 | 10.0 | 13.10 | 0.49 | 45.14 | 6.7 | 10.67 | 0.49 | 49.71 | 20.0 | 9.88 | 0.40 | 48.34 | 12.2 | 11.22 | 0.46 |
| Qwen3.8-Max | 53.88 | 26.7 | 51.40 | 2.64 | 42.47 | 6.7 | 63.33 | 2.11 | 58.82 | 23.3 | 27.53 | 1.43 | 51.72 | 18.9 | 47.42 | 2.06 |
| Kimi K3 | 57.82 | 20.0 | 14.90 | 1.43 | 53.34 | 20.0 | 38.68 | 6.37 | 57.51 | 16.7 | 12.58 | 0.98 | 56.22 | 18.9 | 22.05 | 2.93 |

## 3.4 Benchmark Composition

As summarized in Table 1, ProfessionalBench comprises **435** validated KAC-Unified tasks: **141** in finance, **167** in law, and **127** in healthcare, covering **30** domain-specific subdomains, **127** professional units, and **355** anchor workflows across diverse professional settings. These counts exclude workflow compression intervention variants. The complete taxonomy, including the professional units and anchor workflows covered within each domain, is provided in Appendix F. An audit of five external benchmark releases further supports the taxonomy's external applicability and reveals within-domain coverage differences in Appendix F.4.

## 3.5 Evaluation Protocol

ProfessionalBench evaluates each run through evidence-grounded hybrid verification and reports two metrics: an *Outcome Score* for requirement completion and a binary *Professional Pass* for professional acceptability across diverse real-world professional workflows.

**Evidence-Grounded Hybrid Verification.** After each run, we freeze the agent's workspace and environment state. Each Rubric is evaluated against its designated target in the task-required records, including final artifacts, intermediate work products, and responsibility-related evidence. *Programmatic Verifiers* check deterministic conditions, while an *Agent Verifier* inspects the frozen evidence to assess open-ended professional requirements against expert-reviewed criteria. Each verdict is grounded in supporting evidence and accompanied by a concise rationale.

**Outcome Scoring and Professional Gates.** For task $t$, let $\mathcal{R}_t$ denote ordinary scored rubric items, $w_{t,i}$ their signed weights, and $h_{t,i} \in \{0,1\}$ whether the corresponding predicate is true. Positive-weight items describe desirable properties, whereas negative-weight items describe failure events. Critical requirements are implemented as disqualifying hurdles. Let $\mathcal{H}_t$ index their violation predicates, where $h_{t,j} = 1$ indicates a disqualifying failure. With $g_t = \prod_{j \in \mathcal{H}_t}(1 - h_{t,j})$, we define

$$S_t = g_t \max\left\{0, \frac{\sum_{i \in \mathcal{R}_t} w_{t,i} h_{t,i}}{\sum_{i \in \mathcal{R}_t} \max(w_{t,i}, 0)}\right\}, \qquad \text{Pass}_t = \mathbf{1}\{S_t \geq \tau_t\}. \tag{1}$$

Penalties reduce the normalized score, while any triggered hurdle sets it to zero and prevents Professional Pass. We set $\tau_t = 0.85$ and report Outcome Score as $100S_t$. All weights and hurdles are fixed before evaluation.

# 4 Experiments

## 4.1 Experimental Setup

**Benchmark.** We evaluate an expert-selected, domain-balanced subset of 90 tasks, with 30 tasks each in finance, legal, and healthcare. The subset is sampled from the frozen 435-task benchmark, and its selection and

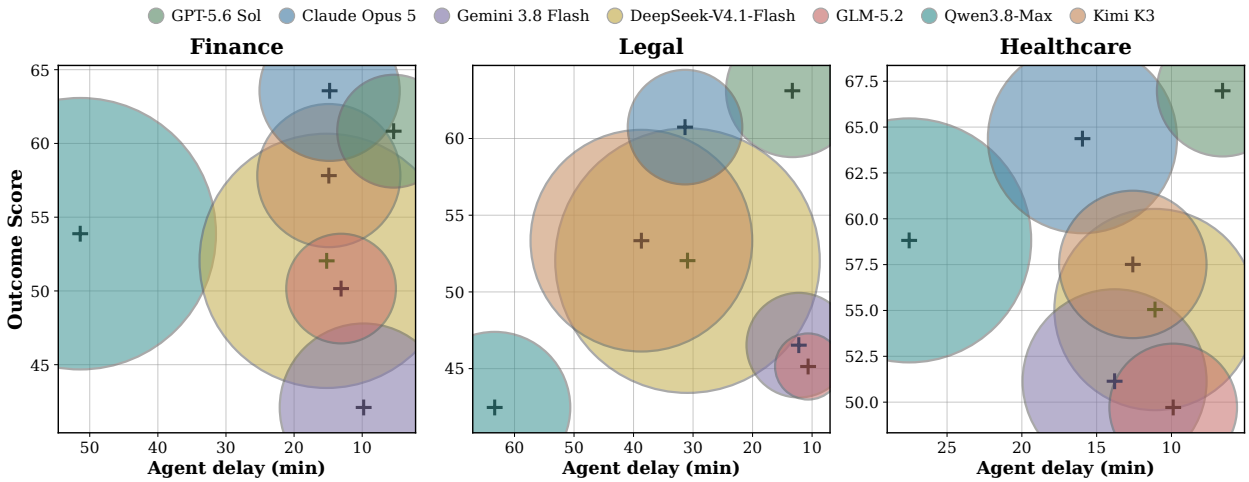



**Figure 3 Model performance and efficiency.** Comparison of model performance, cost, and delay across domains. Bubble size indicates token consumption.

composition are described in Appendix F.5. The tasks cover evidence assessment, business execution, and professional verification with diverse outputs, including reports, spreadsheets, and structured records.

**Models.** We evaluate seven frontier models: GPT-5.6 Sol, Claude Opus 5, Gemini 3.8 Flash, DeepSeek-V4.1-Flash, GLM-5.2, Qwen3.8-Max, and Kimi K3. All models are run through Harbor. Configuration details, including model snapshots, harness versions, tool permissions, and execution environments, are provided in Appendix C.

**Evaluation.** After each run, we freeze the workspace and environment state and provide the evaluator with rubrics, reference results, and verification scripts. Programmatic Verifiers check deterministic conditions, while GPT-5.6 Sol with the Codex harness serves as the Agent Verifier. Verdicts are grounded in the submitted artifacts, execution records, and final business states. We report Outcome Score and Professional Pass following Sec. 3.5 and Appendix B reports the expert-review protocol and agreement results.

## 4.2 Main Results

Table 2 reports performance on ProfessionalBench across finance, law, and healthcare, and Figure 3 summarizes the performance-efficiency trade-offs.

**Frontier agents achieve moderate scores but limited professional acceptance.** GPT-5.6 Sol leads overall in Outcome Score (63.57) and Professional Pass Rate (33.3%), narrowly ahead of Claude Opus 5 (62.94 and 32.2%). Claude leads in finance Outcome Score and healthcare Pass Rate, while GPT performs better in law and healthcare Outcome Scores. Nevertheless, all domain-level Pass Rates remain below 40%, indicating that partial rubric completion rarely translates into professional acceptance, which requires meeting score thresholds and critical requirements.

**Efficiency Analysis.** In terms of cost, token consumption and monetary expenditure exhibit distinct patterns. As shown in Figure 3, DeepSeek-V4.1-Flash and Qwen3.8-Max consume relatively many tokens, yet DeepSeek averages only $0.43 per task and is the least expensive in finance and healthcare, while GLM-5.2 is cheapest in law. However, DeepSeek's overall Outcome Score (53.04) and Pass Rate (14.4%) remain below those of the leading agents, indicating that lower monetary cost does not necessarily coincide with stronger professional performance. In terms of delay, GPT-5.6 Sol achieves substantially lower delay than Claude Opus 5 across all domains, alongside competitive or superior performance. In law and healthcare, GPT attains higher Outcome Scores with shorter execution times; in finance, Claude gains 3.09 Score points but requires approximately 2.7 times the delay. Meanwhile, Qwen3.8-Max exhibits the longest delays across all domains without leading performance. Across the evaluated model–harness configurations, longer execution times are not consistently associated with better professional outcomes.

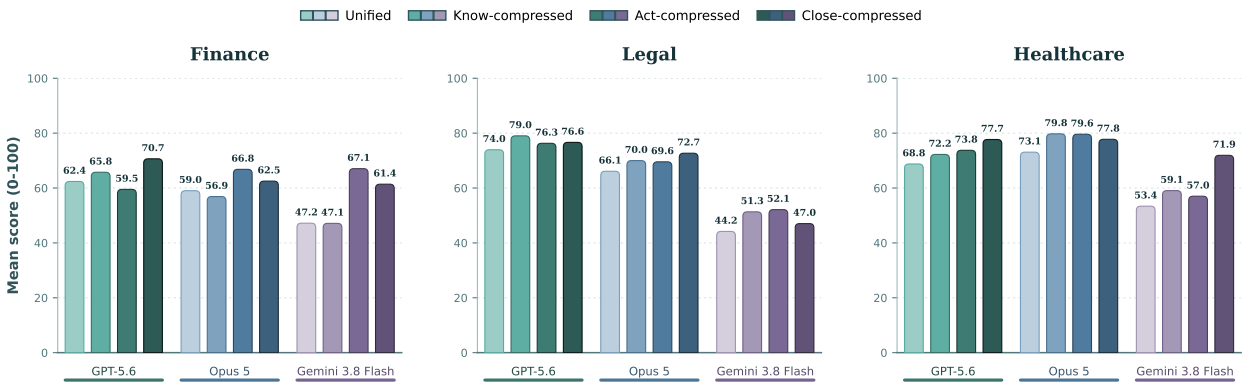



**Figure 4 Workflow compression variants.** Mean Outcome Scores on 30 tasks (10 per domain), evaluated using three agents under Unified, Know-compressed, Act-compressed, and Close-compressed settings. All settings use identical scoring criteria.

# 5 Discussion

## 5.1 Workflow Compression Interventions

We randomly sample 30 tasks from the 90-task evaluation subset, with 10 tasks per domain and without conditioning on baseline scores or Professional Pass outcomes. For each task, we construct Know-, Act-, and Close-compressed variants that supply task-specific intermediate results and pre-resolve selected responsibilities. We evaluate all four settings using the same three frontier agents, yielding 90 matched task–model pairs per variant. The business objective, role boundary, deliverables, execution settings, budget, rubric criteria, weights, normalization, and professional gates remain fixed, including criteria directly supported by the supplied materials. Appendix G.1 details sampling and variant construction.

**Variants' Results.** Figure 4 shows heterogeneous effects across models and domains. Overall, Close compression yields the largest mean Outcome Score increase (+7.80 points), followed by Act (+5.96) and Know (+3.67). The aggregate ordering is not universal: Act yields the largest mean gain for Claude Opus 5, whereas Close does so for GPT-5.6 Sol and Gemini 3.8 Flash.

**Table 3 Paired compression effects.** Changes from Unified over 90 matched pairs (30 tasks across three domains, three agents). All rates use these pairs.

| Metric | Know | Act | Close |
|---|---|---|---|
| Mean score gain (points) | 3.67 | 5.96 | 7.80 |
| Improved trials (%) | 65.56 | 61.11 | 65.56 |
| New passes (%) | 12.22 | 15.56 | 20.00 |
| Lost passes (%) | 0.00 | 0.00 | 0.00 |

**Professional Pass Transitions.** Table 3 shows that Know, Act, and Close compression change 11, 14, and 18 of the 90 paired trials from failing to passing, respectively. No Pass-to-Fail transitions are observed in this sample. The pooled Professional Pass Rate rises from 12.2% under Unified to 24.4%, 27.8%, and 32.2%, demonstrating changes in acceptance decisions under unchanged scoring criteria.

**Analysis.** Compression can change both professional acceptance and the relative ordering of evaluated configurations: under Act compression, Claude Opus 5 overtakes GPT-5.6 Sol in aggregate Outcome Score and Pass Rate on this subset. Yet higher scores need not produce passes: every Gemini law trial gains score under Know compression, while none passes. With rubrics fixed, gains can reflect requirements directly supported by the supplied materials and changes in downstream execution. Higher scores or passes therefore do not establish that agents can independently fulfill the responsibilities pre-resolved by the task package. Appendix G.2 details the aggregation protocol, recovery rates, and additional case studies.

## 5.2 Failure-mode Analysis

Compression interventions measure how supplied intermediate results change performance under fixed acceptance criteria; failure diagnosis complements them by examining Unified runs, where the corresponding

**Table 4 Failure attribution taxonomy.** Categories and types used to classify the primary failure of each task–model trial. Each classified trial contributes one failure type and its corresponding category to aggregate statistics.

| Primary category | Failure type | Core definition |
|---|---|---|
| Know | Domain Knowledge Gap | Missing or misapplied professional knowledge, domain rules, concepts, or judgment. |
| | Evidence Selection / Interpretation Error | Incorrect selection or interpretation of task evidence. |
| | Hallucination / Fabrication | Unsupported facts, data, citations, files, or claimed tool results. |
| Act | Wrong Strategy / Reasoning Error | An unsuitable method, analytical framework, calculation basis, or reasoning path. |
| | Implementation Bug | A local error in code, formulas, calculations, data processing, parameters, or state updates. |
| | Boundary / Safety Violation | Unauthorized actions, unsafe operations, or improper changes to protected files or state. |
| Close | Incomplete / Abandoned | Required steps, checks, content, or deliverables are omitted or left unfinished. |
| | Communication / Format Error | The final output is unclear or violates file, template, schema, precision, or delivery requirements. |
| | Non-actionable Output | Recommendations or plans are present but infeasible or insufficiently specific for execution. |
| Infrastructure | Timeout / Resource Limit | The run exceeds available wall-clock time, tokens, memory, or compute. |
| | Harness / Environment Error | The run fails because of tool, API, connection, environment, or agent-harness faults. |

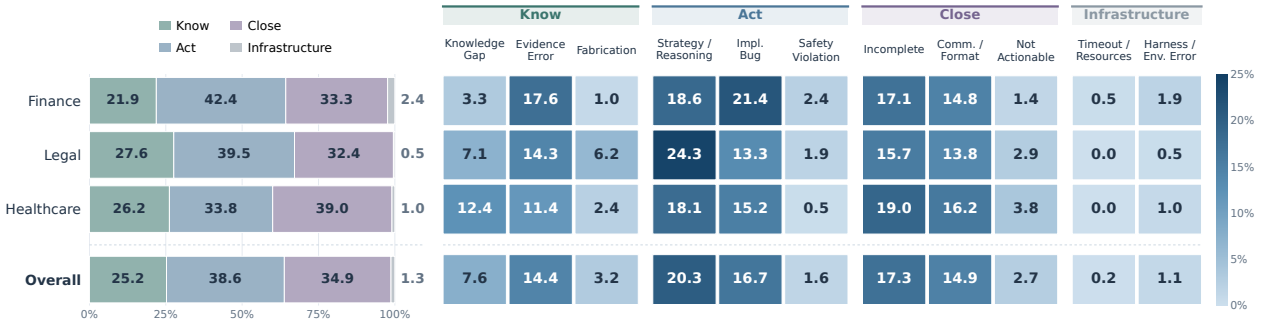



**Figure 5 Primary failure attribution across domains.** Distribution of primary failure categories and types across task–model trials on Unified tasks. Each classified trial contributes one primary diagnosis; secondary failures are excluded. Domain-level percentages are normalized over classified trials within the corresponding domain.

responsibilities remain with the agent. We combine Dense Rubric verdicts with execution trajectories: rubrics locate unmet acceptance conditions, while trajectories support attribution to underlying errors and omissions. Correlated rubric failures are grouped when they share an upstream cause, and each classified task–model trial contributes one primary failure type and its corresponding category. Figure 5 reports the composition of these primary diagnoses rather than per-task failure probabilities.

**Overall.** Act and Close account for 38.6% and 34.9% of primary diagnoses, respectively, together comprising 73.5%; Know contributes 25.2% and Infrastructure 1.3%. The most frequent individual types are strategy or reasoning errors (20.3%), incomplete or abandoned work (17.3%), and implementation bugs (16.7%). These patterns highlight difficulties in selecting suitable approaches, executing them correctly, and carrying required work through to completion.

**Domain-Specific.** Act is the most frequent primary category in finance (42.4%) and law (39.5%), although their leading failure types differ. Finance is led by implementation bugs (21.4%) and strategy or reasoning errors (18.6%), whereas law is led by strategy or reasoning errors (24.3%) and incomplete or abandoned work (15.7%). In healthcare, Close is the most frequent category (39.0%), with incomplete or abandoned work (19.0%) and communication or format errors (16.2%) as its main types. These distributions suggest different diagnostic priorities across domains: implementation correctness in finance, strategy selection in law, and completion and delivery in healthcare.

## 5.3 Representative Case Studies

Figure 6 presents a home-purchase planning task under school-entry and household cash-flow constraints. The agent must assess source and policy applicability, compare financing scenarios, and verify the feasibility and consistency of the purchase plan and supporting work records.

**Compression and Pass Transitions.** Know-compressed supplies guidance on source priority and policy applicability; Act-compressed scaffolds feasibility checks and scenario comparison; Close-compressed provides

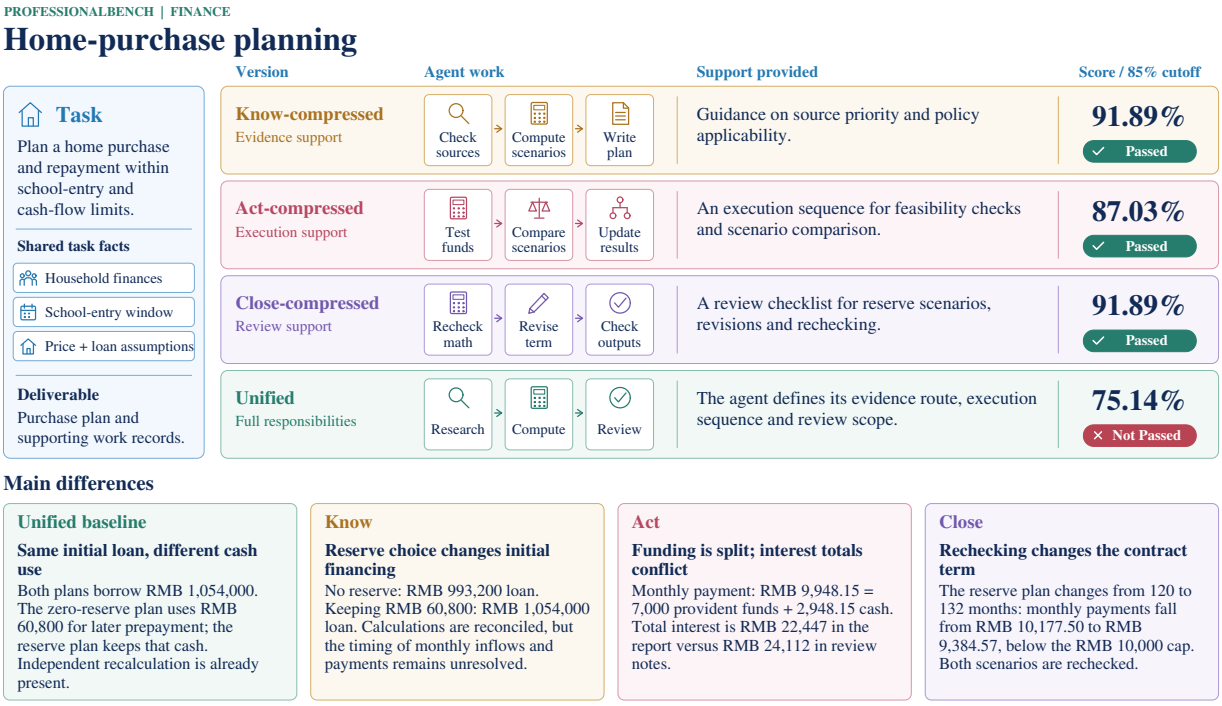



**Figure 6 Professional Pass transitions in a finance task.** The Unified home-purchase plan does not pass, whereas all three compressed variants achieve Professional Pass under unchanged acceptance criteria.

review guidance for reserve scenarios, revisions, and rechecking. Unified scores 75.14 and does not achieve Professional Pass. Scores increase to 91.89, 87.03, and 91.89 under Know, Act, and Close compression, respectively, with all three compressed runs achieving Professional Pass under unchanged acceptance criteria.

**Diagnostic Implications.** Unified already includes recalculation, but the variants differ in how evidence, execution, and review support shape the final plan. In the Close-compressed run, the reserve scenario's loan term changes from 120 to 132 months, reducing the calculated monthly payment from RMB 10,177.50 to RMB 9,384.57, below the RMB 10,000 cap. These observed Fail-to-Pass transitions show that supplying part of the professional work can change acceptance outcomes without establishing stronger independent end-to-end capability. Appendix G.2 provides additional cases with selective Pass transitions and score gains without acceptance.

## 6 Conclusion

We introduced **ProfessionalBench**, a benchmark for evaluating AI agents on real-world professional workflows while preserving the responsibilities required for professional judgment, execution, and verification. By identifying *Workflow Compression* and operationalizing responsibility preservation through *Know–Act–Close*, ProfessionalBench enables more faithful evaluation and fine-grained diagnosis of agent capabilities. Experiments across finance, law, and healthcare show that strong task completion does not necessarily translate into professional acceptance, and that different forms of responsibility compression lead to distinct capability patterns. We hope ProfessionalBench provides a foundation for developing more reliable evaluations of AI agents in complex professional settings.

## 7 Contributions

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# Appendix

## A Professional Task Examples

### A.1 Finance

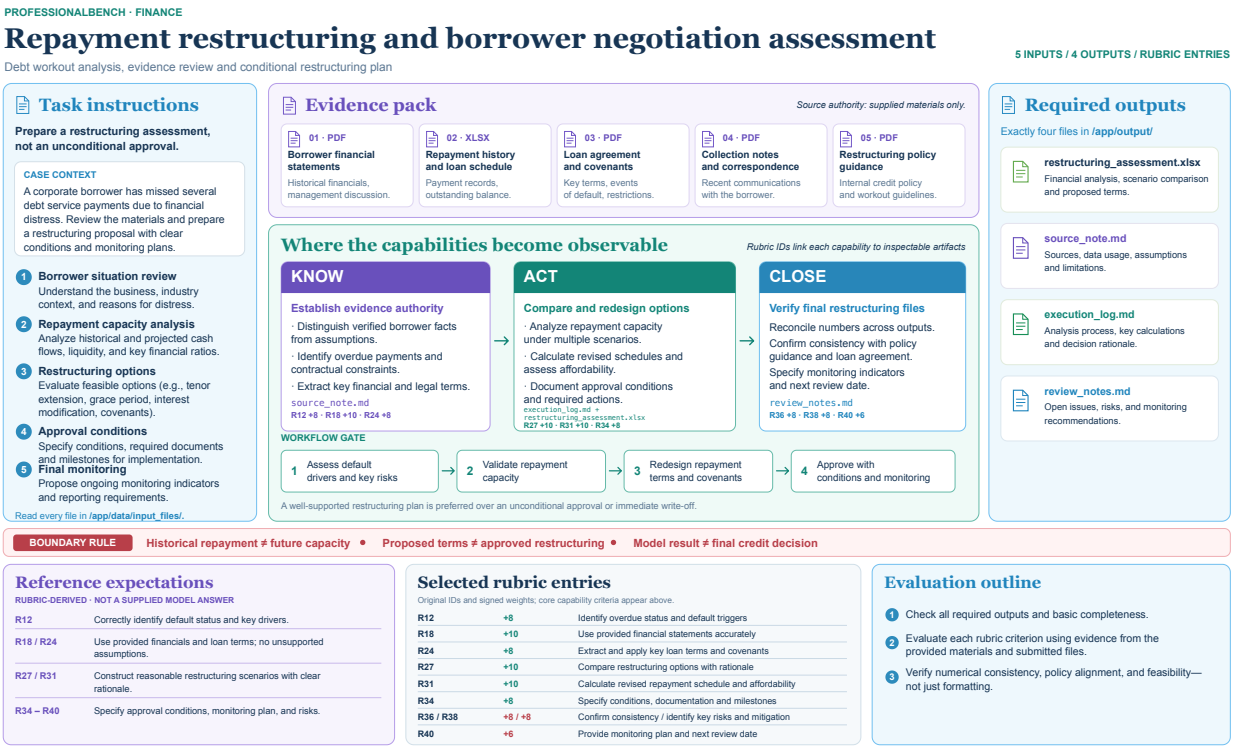



**Figure 7 Finance task example.** A representative finance workflow showing the professional role, task objective, input materials, required deliverables, KAC capability responsibilities, and rubric-based verification criteria.

### A.2 Legal

### A.3 Healthcare

## B Verification Consistency and Reliability

### B.1 Expert Review Setup

**Task and rubric review.** During benchmark construction, domain experts reviewed the task specifications and acceptance rubrics for all 435 tasks (141 in finance, 167 in law, and 127 in healthcare), covering 15,109 rubric items. They assessed professional relevance, responsibility coverage, and the clarity and verifiability of acceptance conditions. The rubrics used for scoring were fixed before model evaluation.

**Verifier review.** Experts reviewed the GPT-5.6 Sol Agent Verifier's run-level Professional Pass decisions for all 630 task–model trials, covering seven evaluated configurations on 90 tasks, with 30 tasks per domain. The review included both automatically accepted and rejected submissions. Experts checked the decisions against the corresponding task materials, submitted artifacts, and supporting execution evidence.

### B.2 Expert Review Results

**Rubric quality.** The final expert review approved 92.1% of rubric items. All items not approved in this review subsequently underwent one round of revision based on expert feedback. The reported approval rate reflects rubric quality before this revision, rather than agreement on model outputs.




**Figure 8 Legal task example.** A representative legal workflow illustrating how factual materials, applicable rules, professional objectives, deliverables, KAC responsibilities, and evidence-based evaluation criteria are structured within a benchmark task.

**Professional Pass agreement.** The original automatic Professional Pass decisions agreed with expert assessments on 610 of 630 trials (96.8%). Agreement was measured before expert correction of the automatic decisions.

# C Implementation Details

## C.1 Task Packages and Execution Environments

ProfessionalBench contains 435 executable task packages across three professional domains. Each package is self-contained and includes task instructions, input materials, execution specifications, hidden references, and verification logic. Table 5 summarizes the benchmark scale.

**Table 5 ProfessionalBench task package statistics.**

| Statistic | Value |
|---|---|
| Total executable tasks | 435 |
| Finance / Law / Healthcare tasks | 141 / 167 / 127 |
| Evaluation subset | 90 tasks (30 per domain) |
| Input files | 8,308 |
| Required deliverables | 1,880 |
| Rubric items | 15,109 |
| Average inputs per task | 19.1 |
| Average deliverables per task | 4.3 |
| Average rubrics per task | 34.7 |

Each task package contains a solver-visible *instruction.md*, input files under *data/input_files/*, a *task.toml*

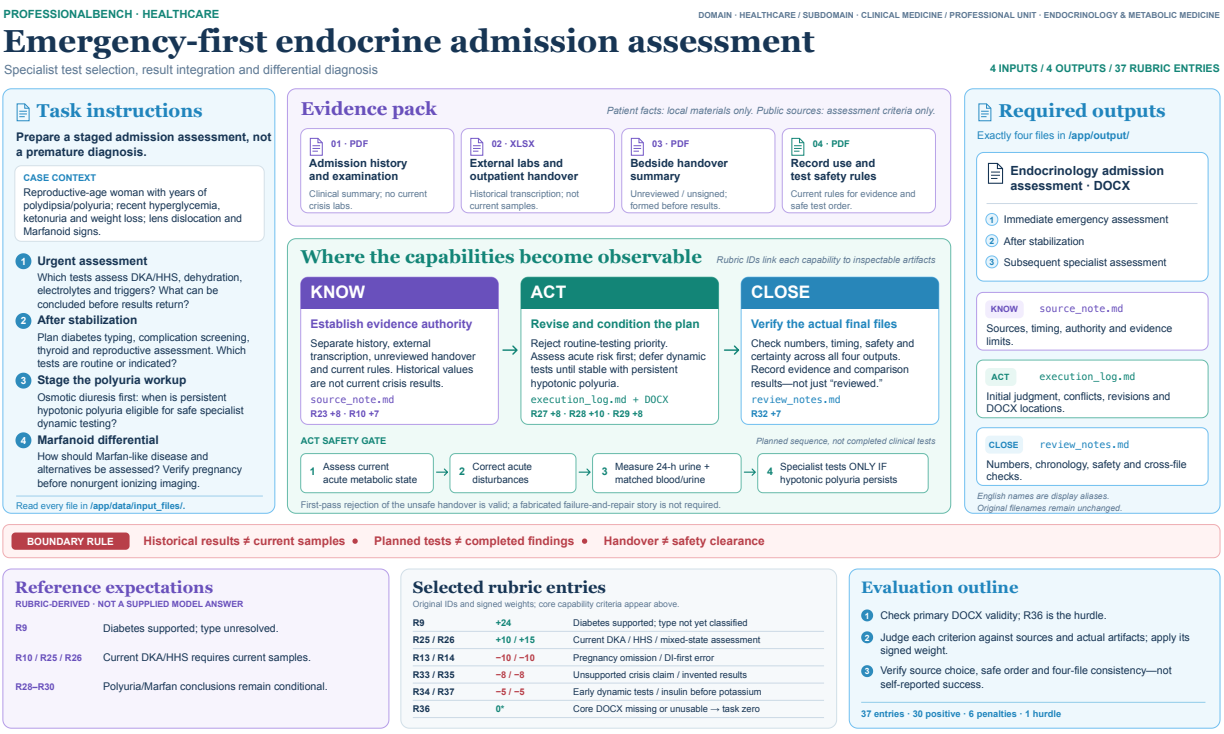



**Figure 9 Healthcare task example.** A representative healthcare workflow demonstrating evidence integration, professional judgment, required outputs, KAC responsibilities, and rubric-based assessment of task completion.

**Table 6 Model and agent harness configurations.**

| Model | Inference Route | Agent Harness |
|---|---|---|
| GPT-5.6 Sol | *azure/gpt-5.6-sol* | Codex CLI 0.154.0 |
| Claude Opus 5 | *anthropic/claude-opus-5* | Claude Code 2.1.272 |
| Gemini 3.8 Flash | *gemini/gemini-3.8-flash* | Gemini CLI 0.60.0 |
| DeepSeek V4.1 Flash | *volcengine/deepseek-v4-1-flash-260910* | Claude Code 2.1.272 |
| GLM-5.2 | *volcengine/glm-5-2-260617* | Claude Code 2.1.272 |
| Kimi K3 | *volcengine/kimi-k3-260716* | Kimi Code 2.0.2 |
| Qwen 3.8 Max | *dashscope/qwen3.8-max* | Qwen Code 0.24.1 |

execution specification, an isolated Docker environment, hidden reference materials, and a *tests/* directory containing verification scripts, *rubric.json*, and optional deterministic checks. Reference solutions are never exposed to the solver and are mounted only during scoring.

We use Harbor 0.22.0 and Docker 29.6.1. Tasks execute in isolated ARM64 Linux containers based on *python:3.12-slim-bookworm*, with task-dependent limits of 1–4 CPUs, 1–8 GiB memory, 2–20 GiB storage, no GPU access, and a 7,200-second agent timeout. Prepared task snapshots, source files, and runtime images are tracked using SHA-256 hashes for reproducibility.

## C.2 Models, Harnesses, and Execution Settings

Models are accessed through the Humanlaya gateway using the routes specified in Harbor configurations. Table 6 lists the evaluated models, inference routes, and agent harnesses.

DeepSeek and GLM used text-only Anthropic-compatible routes. For healthcare and law GLM runs, Claude

Code's *Read* tool was disabled, while Bash, Python, PDF extraction, and OCR remained available.

Each task–model pair was executed once, resulting in $90 \times 7 = 630$ primary trials. Runs used two-way concurrency with at most one infrastructure retry. Solver outputs were evaluated immediately by Codex CLI 0.154.0 using *azure/gpt-5.6-sol*. For criteria assessed by the Agent Verifier, judgments were processed separately with a maximum concurrency of two calls, a 600-second per-criterion timeout, a 7,200-second verifier timeout, and up to three API attempts.

### C.3 Reproduction Instructions

Experiments can be reproduced from *ProfessionalBench_ Evaluation/official_ evaluation*. After installing the validated Harbor version, users should configure the required model providers by specifying API keys, endpoint URLs, model identifiers, and routing names in the provided configuration files. These entries must match the model routes reported in Table 6; otherwise, the resulting runs may not correspond to the reported evaluation setting.

After environment setup, users can audit task packages, execute smoke tests, and launch official runs using the scripts provided in our codebase. The evaluated profiles include *codex-gpt56*, *cc-opus5*, *gemini38-flash*, *cc-deepseek*, *cc-glm*, *kimi-k3*, and *qwen38-code*. The archived Harbor configurations, solver trajectories, generated artifacts, verifier logs, per-rubric *verifier/result.json*, and final *result.json* files are the authoritative sources for the reported scores.

## D Evaluation Details and Dense Rubrics

This section describes the evaluation environments and dense rubric schema, followed by the scoring rules for signed criteria, dependencies, and hard gates.

### D.1 Evaluation Environments Setup

We use two complementary verification methods.

**Deterministic programmatic evaluation.** Criteria with explicit ground truth are evaluated using executable checks. Examples include comparing spreadsheet values with reference calculations, matching required strings, and checking file existence or validity. Comparisons follow predefined rules, including numerical tolerances where applicable.

**Agent-as-a-judge.** For criteria requiring semantic or professional judgment, an agent inspects the response and generated deliverables against the rubric. It may consult source materials or execution logs to verify supporting evidence. Each check produces a binary verdict and a brief evidence-grounded justification. Verifier errors are recorded separately rather than treated as failed criteria.

### D.2 Rubric Schema

A *criterion* is a self-contained, atomic, binary-verifiable predicate. A *rubric entry* combines the criterion with its annotations and scoring metadata, as summarized in Table 7.

**Capability tags.** Rubric entries follow the Know–Act–Close framework: **Know** covers professional knowledge, evidence understanding, and groundedness; **Act** covers strategy, reasoning, implementation, and execution boundaries; and **Close** covers completeness, communication, delivery, and actionability. Each entry receives one primary domain and one corresponding subdimension from Table 4. These capability tags are distinct from the workflow *stage*, which records Read, Process, or Output.

**Failure attribution.** Failure labels are assigned at the task–model trial level using execution evidence and final deliverables, rather than directly converting failed rubric tags into failure types. Each classified trial contributes one primary failure type and its corresponding Know, Act, Close, or Infrastructure category to aggregate statistics. Secondary issues are retained in the diagnostic record but are not counted as additional failures.

**Table 7 Dense rubric schema.** Categorical values are shown in English. Enum fields are single-choice; optional dependency and control fields may be omitted.

| Field | Type | Definition and values |
|---|---|---|
| *id* | Integer | Stable identifier within the task rubric. |
| *criterion* | String | Atomic binary predicate specifying the assessed property and its verdict condition. |
| *stage* | Enum | **Read**: interpret instructions and acquire information; **Process**: integrate, calculate, and reason; **Output**: produce the response or deliverables. |
| *target* | Enum | **Response**: user-facing reply; **Artifact**: generated files; **Trajectory**: observable operations and execution records. |
| *necessity* | Enum | **Explicit**: directly required by the prompt; **Implicit**: necessary for task completion or professional practice; **None**: an auxiliary check not separately assigned to either requirement category. |
| *weight* | Signed integer | Importance-weighted credit or penalty: positive values reward desirable properties; negative values penalize failure events. |
| *dimension* | Enum | Primary capability domain: Know, Act, or Close. |
| *subdimension* | Enum | One capability tag within the selected domain, using the eleven subdimensions in Table 4. |
| *info_source* | String | Evidence location, such as a prompt clause, document page, spreadsheet range, database table, or execution log. |
| *info_source_detail* | String | Specific reference information or justification supporting the check. |
| *depends_on* | Integer | Optional predecessor ID; dependency links must be acyclic. |
| *hurdle* | Boolean | Hard-gate flag: a true predicate forces the task score to zero. Default: *false*. |
| *bouns* | Boolean | Extra-credit flag, tracked separately from the ordinary score. Default: *false*. |

**Table 8 Example rubric entries.** Criteria are abbreviated for presentation. Entry 3 records a dependency; entry 5 is a task-level hard gate.

| ID | Criterion | Weight | Control |
|---|---|---|---|
| 1 | Samantha Foster's adjusted unexcused absence count is $10 - 1 = 9$: deduct the January 14 absence covered by approved leave, but not the January 15 partial absence. | +10 | — |
| 2 | The report provides narrative analysis covering all five departments identified in the supplied staff database. | +6 | — |
| 3 | Employee absence counts in the report agree with the corresponding values in the calculation workbook. | +4 | *depends_on: 1* |
| 4 | The response presents fabricated task-specific facts or sources as established evidence. | −8 | — |
| 5 | Any of the three required deliverables—two PDFs and one XLSX workbook—is missing or cannot be opened. | 0 | *hurdle: true* |
| 6 | The response claims to have checked *greenfield_staff.db*, but a complete execution record contains no supporting access or verification evidence. | −6 | — |

## D.3 Scoring and Dependency Controls

**Signed criteria.** For a fixed task $t$, let $z_i = h_{t,i} \in \{0, 1\}$ denote the criterion verdict in Eq. (1). An ordinary rubric item contributes the unnormalized amount $\Delta_i = w_i z_i$, which enters the normalized task score defined in the main text. Where $w_i$ is its signed weight. Thus, a positive-weight entry describes a desirable property, while a negative-weight entry describes a failure event. Entries marked *bouns* receive separate extra credit rather than contributing to the ordinary score.

**Dependencies.** The *depends_on* links form an acyclic graph connecting related checks. Dependencies do not automatically propagate verdicts: a report and workbook can agree on the same incorrect value, so cross-artifact consistency and numerical correctness remain distinct checks.

**Hard gates.** Critical requirements are enforced through rubric entries marked `hurdle`. Each such entry describes a disqualifying failure: when its predicate is true, the task score is set to zero and Professional Pass is denied. This override applies regardless of accumulated credit or bonuses, so a hurdle may have weight zero. Hurdles implement the critical requirements rather than an additional, separate set of acceptance conditions.

Table 8 illustrates these mechanisms using the attendance-analysis task.

# E Failure Diagnosis Protocol

Expert-defined rubrics are inherently goal-oriented: they encode the professional outcomes and capabilities that experts intend to measure. However, rubric-only analysis mainly reveals *which requirements* were not satisfied and may miss latent execution problems or causal mechanisms behind failures. We therefore introduce a complementary failure diagnosis protocol that combines rubric-level requirement analysis with trajectory-based analysis. The former provides structured failure evidence, while the latter reconstructs how failures emerge during execution.

## E.1 Rubric-Level Requirement Analysis

Given each task × run, we first analyze rubric outcomes using simple aggregation statistics. We summarize unmet criteria, failed Professional Gates, and score losses across Know–Act–Close domains, and capability tags. Positive-weight criteria are considered unmet when their predicates are not satisfied, while negative-weight criteria are considered violated when their failure conditions are triggered.

This step answers *what requirements were not satisfied*, rather than directly identifying the underlying cause. In particular, a failed rubric does not automatically imply a specific failure type: the same unmet criterion may result from a knowledge gap, an execution mistake, or incomplete delivery. Causal interpretation is performed only after trajectory-based analysis.

## E.2 Two-Stage Trajectory Analysis

Trajectory analysis separates factual reconstruction from failure classification. For each task–model trial, local Python scripts assemble a source-indexed evidence package containing task instructions, rubric definitions and verdicts, grading rationales, archived trajectories and tool responses, input materials, final artifacts, and environment logs. Unscored criteria, evaluator failures, and missing analysis evidence are recorded separately rather than automatically attributed to the evaluated agent.

Stage I uses chunked reading followed by synthesis. The model first produces source-referenced notes for each of $N$ evidence blocks, and a subsequent request combines these notes into an **Analysis Card**. The synthesis request receives the notes and references rather than the complete original trajectory. The card records observed deviations, recovery attempts, downstream effects, and affected rubric items; each score-losing item is linked to an observed issue or marked as unexplained. Stage I does not assign final failure categories, and evidence may be omitted during summarization.

Each Analysis Card contains:

- task, model, and run identifiers;
- task requirements and key constraints;
- execution summary and correctly completed actions;
- observed errors, affected rubrics, and artifact status;
- evidence pointers to trajectory steps, files, or grader results;
- confidence and unresolved uncertainties.

It only reconstructs what happened, where deviations occurred, and how later outcomes were affected.

**Stage I Prompt (Analysis Card Generation)**
Analyze each task × run using the available task instructions, rubrics, trajectory logs, grader results, generated artifacts, and execution metadata.
The goal of this stage is evidence-grounded reconstruction, not failure classification. Do not assign labels from the failure taxonomy or infer unobserved internal states. A failed rubric indicates an unmet requirement, not necessarily its underlying cause.
For each run, generate an Analysis Card containing:

(1) **Task Summary**: summarize the objective, key constraints, and required deliverables relevant to the observed outcome;
(2) **Correct Behaviors**: identify important steps, analyses, tool calls, calculations, or artifacts that were completed correctly, with evidence pointers;
(3) **Observed Problems**: describe what went wrong by comparing the expected behavior with the actual execution. For each issue, provide: - expected result or requirement; - observed behavior; - earliest observable deviation; - affected rubric items; - supporting evidence locations;
(4) **Execution and Artifact Status**: summarize final artifact quality, missing outputs, grader results, and relevant trajectory events;
(5) **Uncertainty**: distinguish confirmed errors supported by direct evidence from possible causes that require further validation.
Use concrete evidence references whenever possible, such as trajectory steps, tool calls, artifact locations, rubric IDs, or source sections. When multiple failed rubric items originate from the same upstream mistake, describe the shared cause and downstream effects separately.
Do not classify an error as knowledge, strategy, implementation, or completion failure at this stage. The purpose of the Analysis Card is to provide a standardized evidence interface for a later independent failure classifier.

Stage II performs failure attribution in a separate request using only the Analysis Card and the four categories and eleven failure types in Table 4. It receives neither the Stage I conversation history nor direct access to the original trajectory or artifacts. Both stages use the model `GPT-6-Sol`.

The core decision rule is a **Counterfactual Repair Test**:

> *If all other factors remain unchanged, what is the smallest type of repair required to recover the lost performance?*

This criterion distinguishes failures with similar surface symptoms. For example, an incorrect spreadsheet value may correspond to: (i) *Wrong Strategy* when the calculation logic is inappropriate; (ii) *Implementation Bug* when the method is correct but the formula is wrong; or (iii) *Incomplete / Abandoned* when required cells are never completed.

**Stage II Prompt (Failure Classification)**
You are a failure diagnosis agent. Your task is to classify the primary failure mode of an agent run using only the provided Analysis Card and the predefined failure taxonomy.
Do not access the original trajectory, task files, artifacts, or grader logs. The Analysis Card is the only evidence source.
Use the Counterfactual Repair Test: "If all other factors remain unchanged, what is the minimal type of change required to fix the observed failure?"
When multiple rubric items fail, do not treat every failed rubric as an independent cause. Identify whether several failures are downstream effects of the same upstream problem.
Return a structured JSON output with fields: primary failure, evidence, minimal repair, affected rubrics, secondary failures, uncertainties, and confidence.

## E.3 Integrated Failure Localization and Analysis

The final diagnostic record links the trial's primary failure to unmet professional requirements, supporting trajectory evidence, a minimal repair hypothesis, and affected rubric items. Multiple rubric losses arising from the same upstream problem are recorded as related consequences. Independent secondary issues may remain in the record without contributing additional entries to the aggregate failure distribution.

Local validation scripts check evidence references, rubric IDs, coverage of score-losing items, and potentially duplicated diagnoses. The pipeline produces Analysis Cards, diagnosis records, aggregate summaries, and a queue for human review. These checks establish structural consistency and traceability but do not guarantee semantic correctness. Cases with insufficient or conflicting evidence remain flagged rather than being assigned an unsupported diagnosis.

**Trial-level aggregation.** Each classified task–model trial contributes exactly one primary failure label,

irrespective of its number of failed rubric items or secondary issues. Domain-level percentages are computed over the trials included in that domain's classified sample. The distribution measures the frequency of primary diagnoses, not the share of lost rubric points or the total number of errors within a run. The resulting diagnoses are evidence-supported post-hoc interpretations rather than experimentally validated causal effects.

# F Detailed Taxonomy and Benchmark Statistics

This section describes the taxonomy and composition of the frozen ProfessionalBench release, including coverage across domains, subdomains, and Professional Units, task and workflow distributions, and statistics on input resources, required deliverables, and rubric items. These statistics characterize the benchmark independently of model performance.

## F.1 Full Taxonomy and Instantiated Coverage

Figure 10 presents the professional task hierarchy. Healthcare's broader Professional Unit coverage is largely driven by the detailed Clinical Medicine taxonomy, while finance and law span more subdomains. All categories are counted within domains, with similarly named categories treated as distinct nodes.

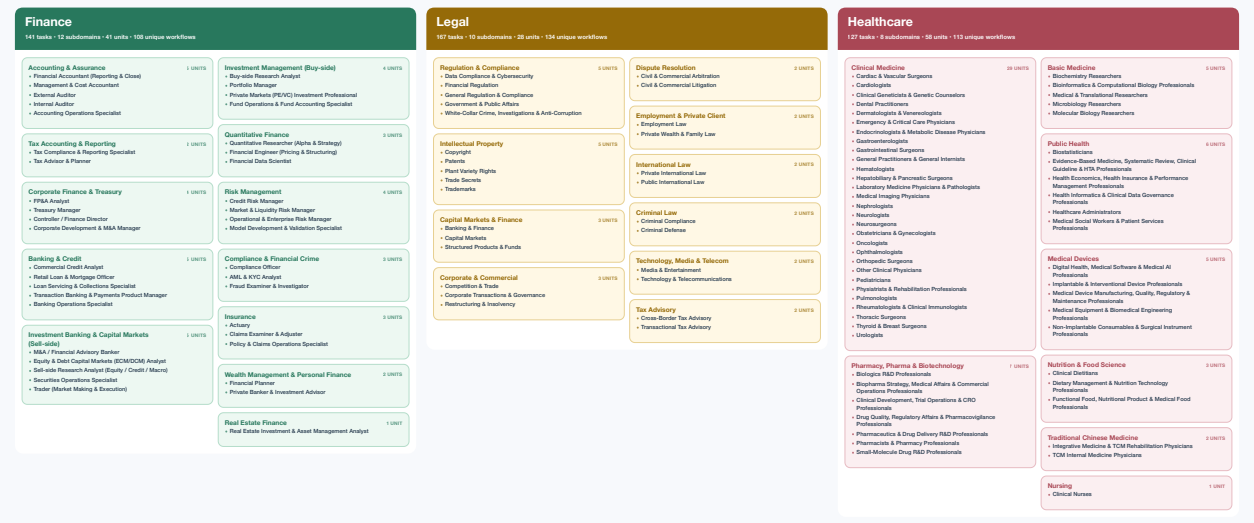



**Figure 10 Full taxonomy and instantiated coverage.** Professional Units organized by domain and subdomain, with domain-level coverage totals.

## F.2 Task and Workflow Distributions

Figure 11 distinguishes coverage breadth from workflow density. Law contains more workflows per Professional Unit, whereas healthcare has broader but sparser unit coverage. The subdomains with the most workflows do not always have the highest density: Investment Management (Buy-side) leads in finance and Dispute Resolution in law when normalized by Professional Unit count.

## F.3 Resources, Deliverables, and Rubric Statistics

Figure 12 shows distinct task-composition profiles. Finance has the most input files and required deliverables per task, while healthcare has the most rubric items per task. Law contributes the largest total numbers of deliverables and rubric items, reflecting its larger task pool rather than the highest per-task counts.

## F.4 External Applicability and Cross-Benchmark Coverage

We apply the ProfessionalBench taxonomy to 13 external benchmark entries, comprising five cross-domain entries and eight domain-specific entries. Table 9 reports the distinct subdomains and Professional Units

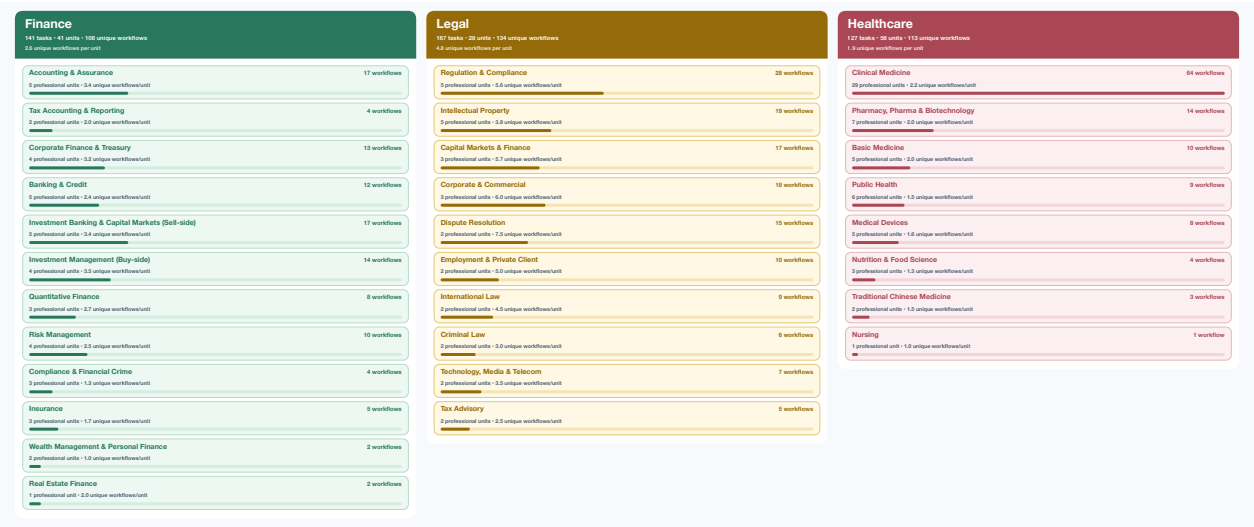



**Figure 11 Task and workflow distributions.** Task totals, distinct Anchor Workflow counts, and workflows per Professional Unit across domains and subdomains.

represented in each audited collection. Cross-domain entries are examined across finance, law, and healthcare, while domain-specific entries are assessed within their target domain. Coverage percentages use the corresponding domain's taxonomy size as the denominator.

**External Applicability.** The five cross-domain collections contain 1,817 audited tasks. The applicability analysis considers the 1,536 tasks identified as falling within finance, law, and healthcare, of which 1,472 (95.8%) can be mapped to the taxonomy. This task-level mapping rate measures applicability to the assessed external tasks; the category counts in Table 9 measure the breadth of professional work represented by each collection.

**Findings and Implications.** Broad subdomain coverage does not ensure complete coverage at the Professional Unit level. PRBench covers every finance and law subdomain but reaches only 33 of 41 finance units (80.5%) and 26 of 28 law units (92.9%). HealthBench Professional has the widest observed healthcare unit coverage, reaching 33 of 58 units (56.9%). These patterns reveal gaps in individual collections relative to the reference taxonomy and motivate selecting new tasks from unrepresented Professional Units. The analysis complements responsibility-preservation auditing: category coverage identifies the types of professional work represented, while the audit examines whether the corresponding responsibilities remain with the agent.

### F.5 Evaluation Subset Selection and Composition

Domain experts reviewed the full collection of 435 tasks and selected 90 representative tasks from commonly encountered subdomains in finance, law, and healthcare, with 30 tasks per domain. The resulting evaluation subset contains 90 tasks, with 30 per domain. As shown in Table 10, it covers 26 of the 30 subdomains and 72 of the 127 Professional Units. The main evaluation results are computed over this subset. We plan to release additional task packages progressively.

## G Workflow Compression Interventions

This appendix describes the construction and paired evaluation of Workflow Compression variants on 30 unified (original) tasks sampled across finance, law, and healthcare.

### G.1 Variant Construction and Evaluation Protocol

**Task selection and design.** We randomly select 10 tasks within each domain of the 90-task evaluation subset, yielding 30 Unified tasks in total. The domain allocation and random seed are fixed in advance; tasks are not

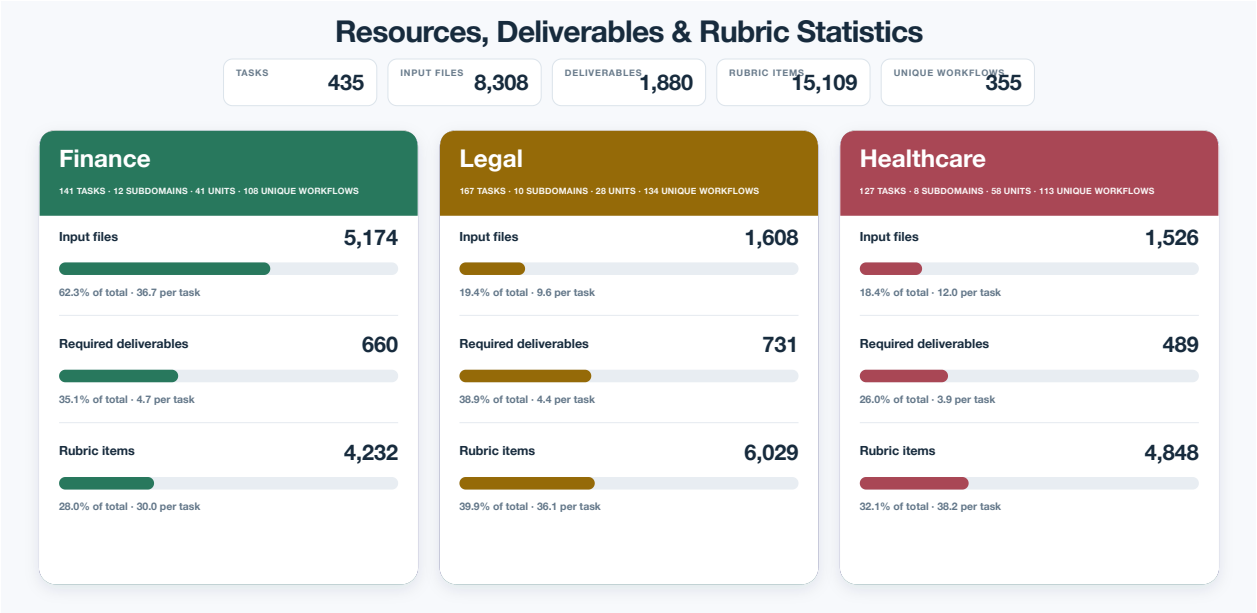



**Figure 12 Resources, deliverables, and rubric statistics.** Input files, required deliverables, and rubric items by domain, with aggregate and per-task statistics.

**Table 9 Cross-benchmark coverage under the ProfessionalBench taxonomy.** Cells report distinct represented categories as counts and percentages, using the domain-specific denominators in the headers. Total units sums unit counts across the reported domains.

| | | Finance | | Law | | Healthcare | | |
|---|---|---|---|---|---|---|---|---|
| **Benchmark** | **Audited tasks** | **Subdomains** (/12) | **Units** (/41) | **Subdomains** (/10) | **Units** (/28) | **Subdomains** (/8) | **Units** (/58) | **Total units** |
| *Cross-domain professional benchmarks* | | | | | | | | |
| GDPval | 220 | 10 (83.3%) | 19 (46.3%) | 3 (30.0%) | 4 (14.3%) | 3 (37.5%) | 6 (10.3%) | 29 |
| APEX-v1-extended | 100 | 9 (75.0%) | 14 (34.1%) | 7 (70.0%) | 8 (28.6%) | 2 (25.0%) | 14 (24.1%) | 36 |
| APEX-Agents-v1.1 | 240 | 11 (91.7%) | 18 (43.9%) | 9 (90.0%) | 15 (53.6%) | 0 (0.0%) | 0 (0.0%) | 33 |
| ALE | 157 | 6 (50.0%) | 8 (19.5%) | 3 (30.0%) | 5 (17.9%) | 5 (62.5%) | 10 (17.2%) | 23 |
| PRBench | 1,100 | 12 (100.0%) | 33 (80.5%) | 10 (100.0%) | 26 (92.9%) | 0 (0.0%) | 0 (0.0%) | 59 |
| *Domain-specific professional benchmarks* | | | | | | | | |
| APEX-Accounting | 10 | 2 (16.7%) | 3 (7.3%) | — | — | — | — | 3 |
| Finance Agent v2 | 27 | 2 (16.7%) | 2 (4.9%) | — | — | — | — | 2 |
| FinGAIA | 150 | 9 (75.0%) | 16 (39.0%) | — | — | — | — | 16 |
| Harvey LAB | 2,010 | — | — | 10 (100.0%) | 25 (89.3%) | — | — | 25 |
| LegalAgentBench | 300 | — | — | 4 (40.0%) | 7 (25.0%) | — | — | 7 |
| MedAgentBench | 300 | — | — | — | — | 2 (25.0%) | 3 (5.2%) | 3 |
| HealthAgentBench | 54 | — | — | — | — | 4 (50.0%) | 5 (8.6%) | 5 |
| HealthBench Professional | 525 | — | — | — | — | 5 (62.5%) | 33 (56.9%) | 33 |

selected or replaced based on baseline scores, Professional Pass outcomes, or compression gains. The same tasks are evaluated using GPT-5.6 Sol, Claude Opus 5, and Gemini 3.8 Flash. We construct 90 compressed task packages, covering three variants per task. Each variant provides 90 task–model comparisons against the same Unified baselines.

**Construction and evaluation.** Experts prepare component-specific materials, including evidence assessments

**Table 10 Composition of the evaluation subset.** Coverage fractions count distinct subdomains or Professional Units represented in the evaluation subset relative to the full task collection.

| Domain | Full tasks | Eval. tasks | Subdomains | Prof. Units |
|---|---|---|---|---|
| Finance | 141 | 30 | 12/12 | 27/41 |
| Law | 167 | 30 | 9/10 | 21/28 |
| Healthcare | 127 | 30 | 5/8 | 24/58 |
| **Total** | **435** | **90** | **26/30** | **72/127** |

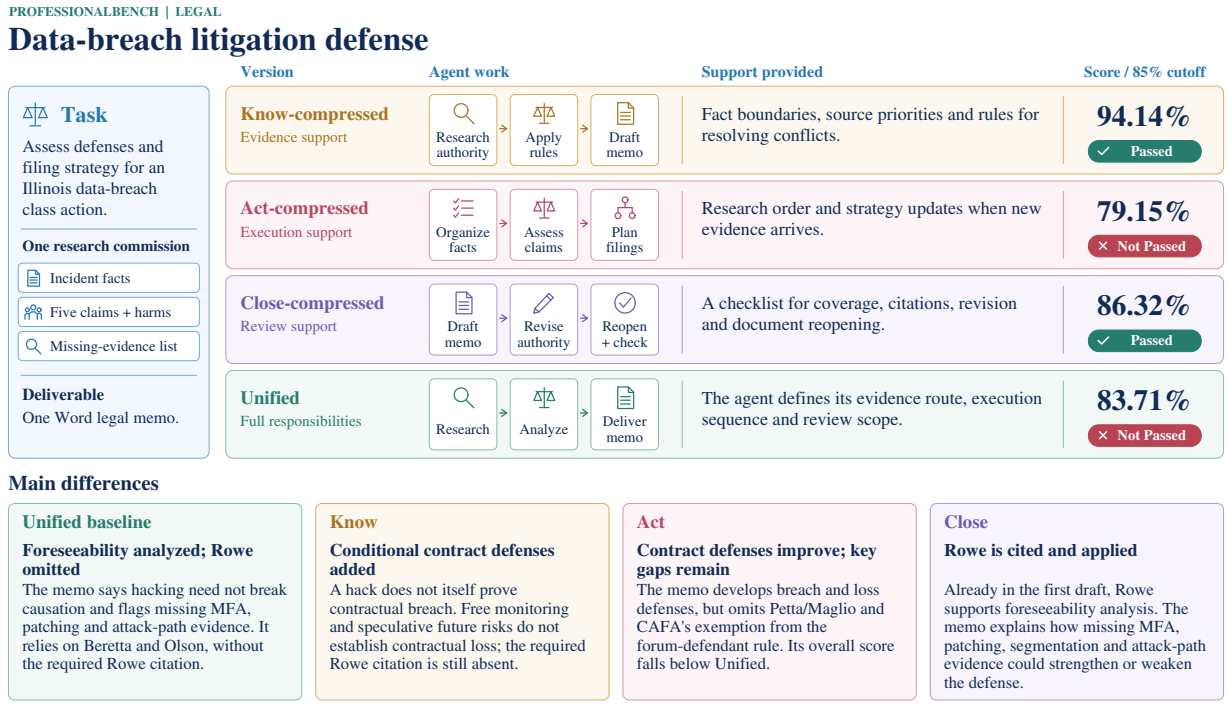



**Figure 13 Case Study in Legal.** Know and Close compression change the illustrated outcome from Not Passed to Passed, while Act compression lowers the score and remains Not Passed.

for Know, plans or execution results for Act, and verification or completion aids for Close. The technical team incorporates these materials into the task instructions and input files. Within each task–model comparison, the objective, required deliverables, execution settings, and budget remain unchanged, as do the complete rubrics, weights, scoring rules, and Professional Pass conditions. Keeping the rubrics fixed allows us to capture requirements that become satisfied through either direct support from the supplied materials or easier downstream work.

## G.2 Paired Results and Case Studies

**Paired Score and Acceptance Changes.** We compare each compressed run with the original run for the same task and model. Each variant is evaluated on $N = 90$ matched task–model pairs, comprising 11 original passes and $N_{\text{fail}} = 79$ original failures. Table 3 reports the mean paired score difference, the proportion of pairs with a strictly increased score, and the Fail-to-Pass proportion. Both proportions use all $N$ pairs as the denominator, including ties and score decreases. Score comparisons use unrounded values, and Professional Pass follows the complete score-threshold and critical-requirement conditions.

Know, Act, and Close compression produce 11, 14, and 18 Fail-to-Pass transitions, respectively, corresponding to 12.2%, 15.6%, and 20.0% of all pairs. Among the 79 original failures, the corresponding recovery rates are

13.9%, 17.7%, and 22.8%. No Pass-to-Fail transitions are observed, and the overall Professional Pass Rate rises from 12.2% to 24.4%, 27.8%, and 32.2%, respectively. Nevertheless, scores decrease in 23, 27, and 22 pairs, showing that preserved binary acceptance does not imply uniformly non-decreasing scores.

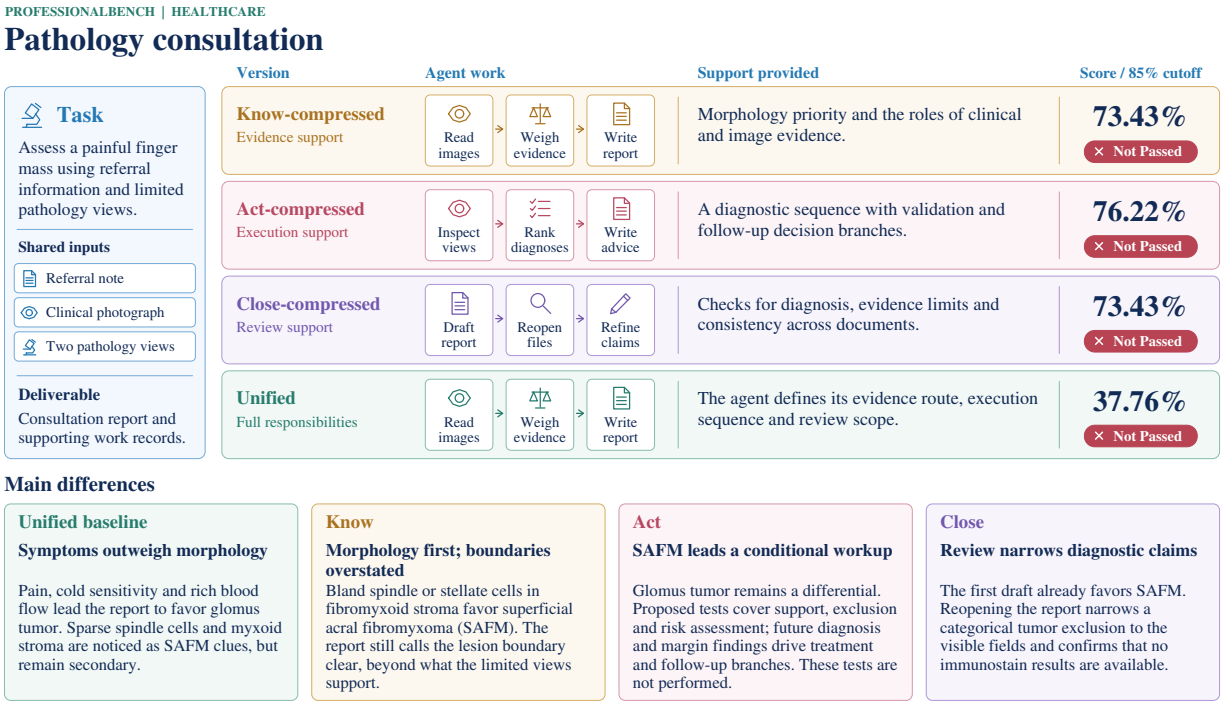



**Figure 14 Case Study in Healthcare.** All three compressed variants substantially increase the Outcome Score, but none achieves Professional Pass.

The finance example in Section 5.3 shows Fail-to-Pass transitions under all three variants. The following cases illustrate two additional responses: acceptance changes under only some variants, and substantial score gains without professional acceptance.

**Law: Selective Pass Transitions.** The data-breach litigation task in Figure 13 requires a legal memo assessing defenses and filing strategy. Unified scores 83.71 and does not pass. Know- and Close-compressed achieve Professional Pass at 94.14 and 86.32, respectively, while Act-compressed falls to 79.15 and remains unsuccessful. Know develops conditional contract defenses, while Close cites and applies the Rowe authority omitted from Unified. Act develops some contract arguments but retains omissions in supporting authorities and filing analysis. The Close variant's 2.61-point gain accompanies a change in professional acceptance, while the Act result shows that compression need not improve every individual run.

**Healthcare: Score Gains without Acceptance.** The pathology consultation task in Figure 14 requires assessment of a painful finger mass from referral information and limited pathology views. Scores rise from 37.76 in Unified to 73.43, 76.22, and 73.43 under Know, Act, and Close compression, respectively. Know places greater emphasis on morphology, Act structures a proposed diagnostic workup, and Close narrows diagnostic claims to the available evidence. Nevertheless, the Know report still overstates what the limited views establish about lesion boundaries, and all three variants remain below the score threshold of 85. Gains of 35.67–38.46 points therefore coexist with continued failure to achieve Professional Pass.

## H Responsibility Preservation Audit across Benchmarks

## H.1 Audit Protocol

We audit tasks from GDPval, APEX-Agents-v1.1, ALE, PRBench, and ProfessionalBench to assess whether material professional responsibilities remain with the solver. The audit examines task construction rather than solver performance. We first identify responsibilities implied by the designated role, task objective, and completion requirements, using source-workflow information where available. These responsibilities are neither restricted to duties explicitly named in the prompt nor extended beyond the task's professional scope. All benchmarks are assessed using the same criteria and evidence-access rules.

**Table 11 Shared criteria for responsibility-preservation auditing.** The auditing agent and human experts examine the same task-specific responsibilities and check both supporting and conflicting evidence. Activities are assessed only where applicable to the task's role and objective.

| Axis | Responsibilities examined | Evidence required for preservation |
|---|---|---|
| **Know** | Identify and assess relevant evidence, determine its applicability, resolve conflicts, and form grounded judgments. | Applicable source-selection and evidence-assessment decisions remain for the solver; supplied materials do not already settle the required judgments. |
| **Act** | Select methods, formulate and execute plans, and adapt to intermediate results where required. | Task-specific decisions and operations remain for the solver; supplied plans or results do not substitute for the required work. |
| **Close** | Verify the actual deliverable, check consistency, perform necessary recalculation, and substantiate completion. | Applicable checks and completion judgments remain for the solver; they are not replaced by already-completed verification results. |

**Positive identification and bidirectional checking.** For each identified responsibility, the auditor first locates evidence that the solver must independently perform the corresponding work, then checks whether instructions or supplied materials instead waive or pre-resolve it. Each responsibility receives a label of *retained*, *omitted*, *pre-resolved*, *not applicable*, or *insufficient evidence*. An axis is confirmed retained only when all its applicable material responsibilities are supported as retained. Retaining some work does not offset a verified omission elsewhere in the same axis. Silence alone does not establish omission, while generic instructions or KAC labels alone do not establish preservation.

**Agent-based assessment.** The auditing agent reads task instructions, conversation history, available attachments, and solver-visible environment descriptions. It records expected responsibilities, preliminary labels, source citations, and missing evidence. Contrastive few-shot examples distinguish substantive professional work from ordinary inputs, blank templates, and general guidance. For a suspected shortcut, the agent explains what work the solver would need to perform if the shortcut were removed while preserving the objective. Automated checks validate references and label consistency, followed by a second-pass check for counterevidence. Uncertain judgments remain unresolved rather than being forced into retained or compressed labels.

**Expert verification.** Domain experts reviewed every task in the audited collections: 157 from ALE, 240 from APEX-Agents-v1.1, 220 from GDPval, 220 from PRBench, and 90 from ProfessionalBench. For each task, experts examined the materials and the agent's supporting evidence to confirm or revise the responsibility labels using the same KAC criteria. Judgments that could not be resolved remained explicitly uncertain. The expert-reviewed results are calculated directly from these final labels, without sampling weights.

## H.2 Audit Results

Table 12 compares agent judgments and expert-verified results for five audited task collections, including 90 Unified tasks from ProfessionalBench. For each axis, retention is confirmed when the axis is applicable and all identified material responsibilities remain with the solver. *All KAC retained* requires this condition to hold jointly for Know, Act, and Close on the same task. *Hard to judge* identifies tasks with unresolved

responsibility or applicability judgments on at least one axis. All percentages use the corresponding collection size as the denominator.

**Table 12 Responsibility-preservation audit results.** Panel (a) reports initial agent judgments; panel (b) reports results after expert verification for the same task collections. Entries give the percentages of tasks with confirmed retention or unresolved judgments.

| Benchmark | Tasks | All KAC retained (%) | Know retained (%) | Act retained (%) | Close retained (%) | Hard to judge (%) |
|---|---|---|---|---|---|---|
| *(a) Agent audit* | | | | | | |
| ALE | 157 | 23.6 | 35.0 | 37.6 | 36.3 | 51.6 |
| APEX-Agents-v1.1 | 240 | 6.7 | 42.9 | 13.8 | 28.3 | 2.1 |
| GDPval | 220 | 42.3 | 60.9 | 79.5 | 69.5 | 0.9 |
| PRBench | 220 | 30.9 | 82.7 | 74.1 | 45.9 | 0.0 |
| **ProfessionalBench** | **90** | **95.6** | **100.0** | **95.6** | **100.0** | **0.0** |
| *(b) Expert verification* | | | | | | |
| ALE | 157 | 26.8 | 54.1 | 49.7 | 52.9 | 21.0 |
| APEX-Agents-v1.1 | 240 | 6.7 | 45.4 | 15.0 | 30.0 | 0.0 |
| GDPval | 220 | 45.0 | 63.2 | 82.3 | 71.8 | 0.9 |
| PRBench | 220 | 33.2 | 84.5 | 75.9 | 48.2 | 0.0 |
| **ProfessionalBench** | **90** | **100.0** | **100.0** | **100.0** | **100.0** | **0.0** |

**Audit Findings.** Expert verification confirms joint KAC retention in all 90 ProfessionalBench-Unified tasks (100.0%), compared with 86 tasks (95.6%) in the agent audit. For the external collections, expert-verified joint retention is 45.0% for GDPval, 33.2% for PRBench, 26.8% for ALE, and 6.7% for APEX-Agents-v1.1. The ordering by joint retention is unchanged after expert review. ALE shows the largest reduction in unresolved judgments, from 81 tasks (51.6%) to 33 tasks (21.0%), although its unresolved rate remains the highest among the audited collections.

**Interpretation.** The distinction between individual-axis and joint retention remains substantial after expert verification. For example, PRBench has Know retention of 84.5% and Act retention of 75.9%, but Close retention of 48.2% and joint retention of 33.2%. High retention on individual axes therefore does not establish that all three responsibility dimensions remain with the solver within the same task. These comparisons characterize the audited collections under the stated responsibility criteria. Unconfirmed retention may reflect omitted or pre-resolved responsibilities, insufficient evidence, or inapplicability, so retention rates should be interpreted alongside task scope and unresolved judgments.

## H.3 Audited Examples

These examples illustrate how task inputs can pre-resolve selected Act and Know responsibilities. Each figure identifies the supplied information, the affected work, and a possible restoration.

**GDPval: Precomputed Aggregates.** The input workbook supplies mean resolution time (23.02 days) and total resolution cost (82,567), pre-resolving two Act computations. Removing the summary formulas and cached results would restore these computations while retaining the incident records and presentation requirements. Other analyses and verification remain with the agent.

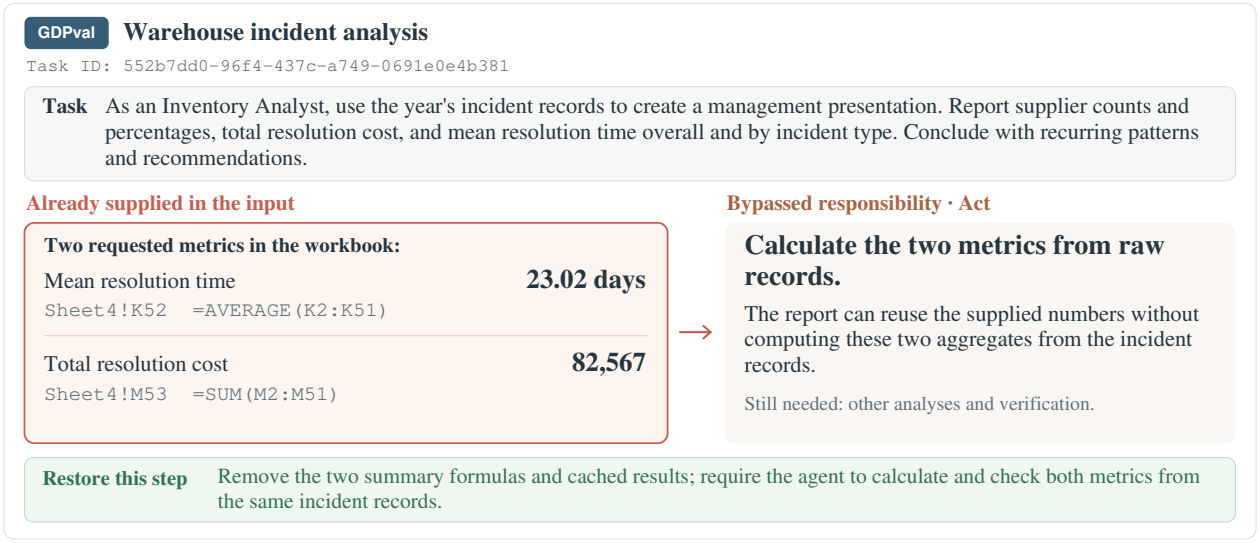



**Figure 15 Act Compression in GDPval.** The input workbook supplies two requested aggregate results that can be reused without independent computation. Other analyses and verification remain with the agent.

**ALE: Prescribed Test Strategies.** The generator task prescribes adversarial input patterns, reducing the Act responsibility of selecting test strategies. Implementation, compilation, and validation remain required. Removing these prescriptions while retaining the problem definition and validity constraints would restore strategy selection without changing the generator deliverable.

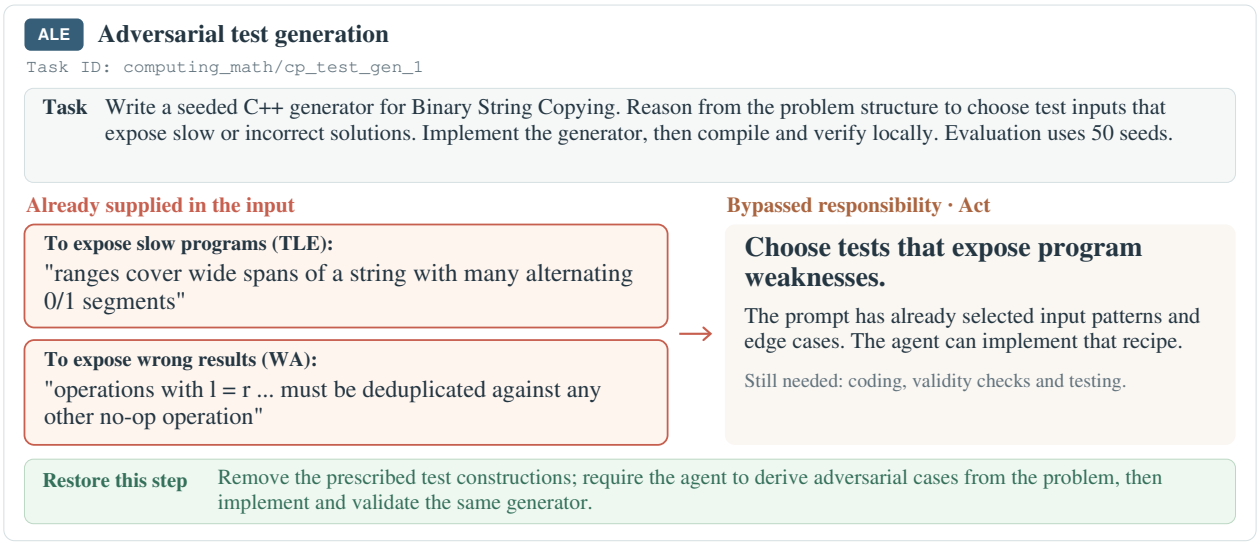



**Figure 16 Act Compression in ALE.** Prescribed adversarial patterns pre-resolve part of test-strategy selection, while generator implementation, compilation, and validation remain required.

**ALE: Supplied Functional Conclusions.** The annotation task states gamma-tubulin family membership and the required functional conclusions in advance, pre-resolving part of Know. InterProScan execution and evidence checks remain necessary. Removing the named conclusions would restore inference from domain and GO evidence while retaining the protein sequence, tools, and deliverables.

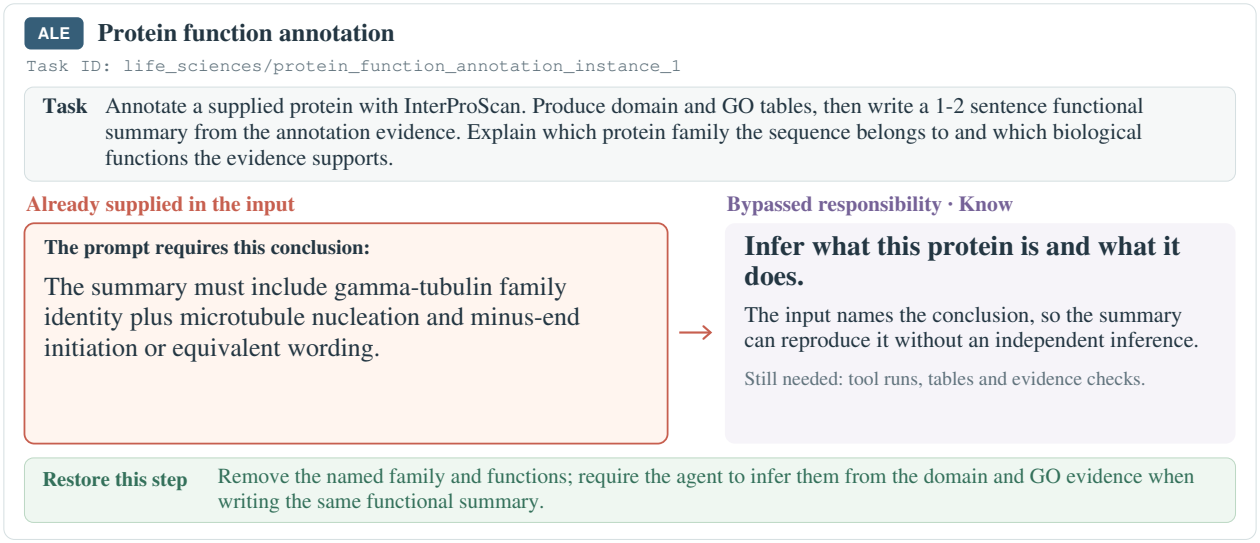



**Figure 17 Know Compression in ALE.** The prompt supplies the expected protein family and functional conclusions, reducing independent inference from annotation evidence. Tool execution and evidence checks remain required.